

MURALI KRISHNA N

Data Scientist | GenAI Engineer | Agentic AI Developer

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PROFESSIONAL SUMMARY

Data Scientist and GenAI Engineer with 1.5+ years of hands-on experience building production-grade AI solutions using Python and Azure. Proven expertise in RAG pipelines, agentic AI workflows (LangGraph), and LLM evaluation frameworks. Delivered client-facing AI projects across Airlines, Retail, and Pharmaceutical domains. Seeking a mid-level role focused on scalable AI systems, agent orchestration, and MLOps best practices.

TECHNICAL SKILLS

Programming: Python, SQL, Git

Machine Learning: scikit-learn, XGBoost, LightGBM, Classification, Regression, Feature Engineering, AUC, F1, RMSE, Class Imbalance Handling, SHAP

NLP and LLMs: Prompt Engineering, Token Limits, Context Windows, Structured Outputs (JSON), LLM Evaluation

Generative AI: RAG Pipelines, Embeddings, Hybrid Retrieval, Chunking Strategies, Citations, Guardrails, Fallback Logic

Agentic AI: LangGraph, LangChain, LlamaIndex, Tool/Function Calling, ReAct Workflows, Planning and Routing, Memory Patterns, Audit Logs

Frameworks: FastAPI, Flask, Pydantic

Vector Databases: Azure AI Search (Vector), FAISS, Chroma

Cloud and Platforms: Azure OpenAI, Azure ML, Azure Blob Storage, ADLS, Key Vault, App Service

MLOps and Evaluation: MLflow, Prompt Versioning, Batch Scoring Pipelines, Data Drift Monitoring, Experiment Tracking

Data and Analytics: Pandas, NumPy, EDA, Matplotlib, Seaborn, Plotly, Data Quality Checks, Feature Engineering

PROFESSIONAL EXPERIENCE

Data Scientist (Intern to Associate)

AI ML LABS PVT LTD, Bengaluru, India | Mar 2025 - Present | 1.5 Years

Delivered end-to-end AI/ML projects for global clients across Airlines, Retail, and Pharmaceutical domains.

- Designed and deployed RAG pipelines on Azure OpenAI and Azure AI Search; reduced airline policy lookup time with grounded, citation-backed responses and confidence-based fallback logic.
- Built a multi-tool LangGraph agentic system for e-commerce returns triage; implemented JSON schema validation, conversation memory, and traceable audit logs for compliance.
- Developed NLP-based adverse event classification model (XGBoost + SHAP) achieving strong F1/recall on imbalanced pharmaceutical datasets; delivered explainability reports for regulatory review.
- Cleaned and prepared structured datasets across projects; handled missing values, outliers, and feature encoding to improve model input quality.
- Tracked model experiments in MLflow, implemented batch scoring pipelines with data drift monitoring, and maintained reproducible notebooks for team handoffs.

PROJECTS

GenAI Operations and Policy Assistant (RAG) - Airlines | AeroLink Airways (APAC)

Role: GenAI Engineer | Stack: Python, Azure OpenAI, LangChain, Azure AI Search (Vector), Embeddings, Prompt Engineering

- Ingested and indexed airline operational manuals using optimised chunking strategies for procedural and tabular data.
- Designed prompt templates generating concise, citation-backed responses with confidence-based fallback logic; optimised cost and latency via targeted prompt tuning.
- Implemented hybrid retrieval with metadata filters to boost domain-specific search accuracy.
- Created scenario-based evaluation datasets covering IRROPS, baggage, and rebooking workflows.

Agentic AI for Returns and Order Exception Resolution - Retail | EComKart Retail (APAC)

Role: Agentic AI Developer | Stack: Python, Azure OpenAI, LangGraph, LangChain, REST APIs, JSON Schema

- Developed a router agent that detects user intent and selects tools (order lookup, policy validation) using strict JSON schema validation for reliable structured outputs.
- Added policy verification before recommending refunds or replacements; built error handling and escalation flows for tool failures and incomplete data.
- Implemented conversation memory and traceable audit logs to support compliance and review requirements.

Adverse Event Seriousness Classification - Pharmaceutical | NovaPharm (US)

Role: Data Scientist, NLP / ML | Stack: Python, scikit-learn, XGBoost, SQL, MLflow, SHAP

- Built supervised ML models using text features and structured case attributes; applied stratified validation with F1 and recall performance tracking.
- Conducted error analysis to improve minority-class detection and reduce false negatives on highly imbalanced data.
- Delivered SHAP-based explainability reports for regulatory reviewers; implemented batch scoring pipelines with data drift monitoring.

EDUCATION

Bachelor of Engineering - Artificial Intelligence and Machine Learning

Jyothy Institute of Technology, Bengaluru, Karnataka, India | Visvesvaraya Technological University (VTU) | 2021 - 2025 | CGPA: 8.1 / 10